Consultații algebră

Spații vectoriale

V_K – $v_1, v_2, \dots, v_n \in V \rightarrow \alpha_1 v_1 + \dots + \alpha_n v_n \in V$ → combinații liniară

Într-un sp vectorial → doar adunarea vectorilor
→ produs scalar

Mulțimea tuturor comb. liniare $\{\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n \mid \alpha_i \in K\}$

$= Sp\{v_1, v_2, \dots, v_n\}$

Dacă $Sp\{v_1, v_2, \dots, v_n\} = V \Rightarrow \{v_1, v_2, \dots, v_n\}$ → sistem de generatori pt. V

(Ex) $\mathbb{R}^3$ $B = \left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$ → s. gen. pt $\mathbb{R}^3$

$$\begin{pmatrix}a\\b\\c\end{pmatrix} = a\cdot\begin{pmatrix}1\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\1\\0\end{pmatrix} + c\begin{pmatrix}0\\0\\1\end{pmatrix}$$

Dependență / Independență liniară

(T) Spunem că $v_1, v_2, \dots, v_n$ sunt liniar dependenți $\Leftrightarrow \exists \alpha_1, \alpha_2, \dots, \alpha_n$ (NU TOȚI NULI) a.î. $\boxed{\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0}$

(T) Spunem că $v_1, v_2, \dots, v_n$ sunt liniar independenți $\Leftrightarrow \forall \alpha_1, \alpha_2, \dots$ α_n și $\alpha_i = 0 \ \forall i$

(Ex) $\mathbb{R}_2[X]$ → mulțt. polin.

↓ $\{-1+X, 2+X, 3-X+X^2\}$

Stud. dependența / indep. liniară.

$$\alpha_1(-1+X) + \alpha_2(2+X) + \alpha_3(3-X+X^2) = 0$$

$$(-\alpha_1+2\alpha_2+3\alpha_3) + X(\alpha_1+\alpha_2-\alpha_3) + \alpha_3 X^2 = 0$$

Trebuie ca toți coef. să fie 0

$$\begin{cases} -\alpha_1+2\alpha_2+3\alpha_3=0 \\ \alpha_1+\alpha_2=0 \\ \alpha_3=0 \end{cases} \Rightarrow \alpha_1=\alpha_2=\alpha_3=0$$

(Ex) $\mathcal{M}_2(\mathbb{R})$ → spațiul matricilor pătratice

↓ $\left\{ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} \right\}$

$$\alpha_1\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + \alpha_2\begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix} + \alpha_3\begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} \alpha_1+2\alpha_2 & \alpha_1-\alpha_2+\alpha_3 \\ 3\alpha_2+2\alpha_3 & \alpha_1-\alpha_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

⇓

$$\left\{ \begin{array}{l} \alpha_1+2\alpha_2=0 \Rightarrow 2\alpha_2=0 \Rightarrow \alpha_2=0=\alpha_3 \\ \alpha_1-\alpha_2+\alpha_3=0 \Rightarrow \alpha_1=0 \\ 3\alpha_2+2\alpha_3=0 \\ \alpha_1-\alpha_3=0 \Rightarrow \alpha_1=\alpha_3 \end{array} \right\} \Rightarrow \alpha_1=\alpha_2=\alpha_3=0$$

⇒ Vectorii liniar independenți

(T) $\mathbb{R}^m$. $\{v_1, v_2, \dots, v_n\}$

$m > n \Rightarrow \{v_1, v_2, \dots, v_n\}$ → liniar dependenți

(Ex) Arătați că e^t și e^{2t} sunt f. l. ni dep. pe $C^\infty(\mathbb{R})$

↓

sp. f. continue derivabile de „∞" ori

$$\begin{cases} \alpha_1 e^t + \alpha_2 e^{2t} = 0 \\ \alpha_1 e^t + 2\alpha_2 e^{2t} = 0 \end{cases} \rightarrow$$ Trebuie să arăt că doar pt. acele val. ale lui t

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$\alpha_2 e^{2t} = 0, \forall t \Rightarrow \boxed{\alpha_2 = 0}$

$\alpha_1 e^t = 0, \forall t \Rightarrow \boxed{\alpha_1 = 0}$

(Ex) $\{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx, \dots\}$ l.i. în $C^\infty(\mathbb{R})$

! Orice mulțime ce conține vectorul nul este liniar dependentă.

(T) Orice mulțime finită este l.d. / l.i. ($\Leftrightarrow$) o submulțime finită a sa este l.d. / l.i.

Fie $f_1, f_2, \dots, f_n \in \mathcal{A}$.

$\alpha_1 f_1 + \alpha_2 f_2 + \dots + \alpha_n f_n = 0$

$\Downarrow$

$\int_{-\pi}^{\pi} \sin mx \cos nx \, dx = 0, \forall m, n$

$\int_{-\pi}^{\pi} \sin mx \cdot \sin nx \, dx = 0, \forall m \neq n$

$\int_{-\pi}^{\pi} \cos nx \cdot \cos mx \, dx = 0, \forall n \neq m$

$\alpha_1 f_1 + \alpha_2 f_2 + \dots + \alpha_n f_n = 0 \mid \cdot f_m$

$\int_{-\pi}^{\pi} \mid \alpha_1 f_1 f_m + \alpha_2 f_2 f_m + \dots + \alpha_n f_n f_m = 0$

$\alpha_1 \int_{-\pi}^{\pi} f_1(x) f_m(x) dx + \alpha_2 \int_{-\pi}^{\pi} f_2(x) f_m(x) dx + \dots + \alpha_n \int_{-\pi}^{\pi} \underbrace{(f_m(x))^2}\, f_n(x) f_m(x) dx = 0$

$\alpha_m \int_{-\pi}^{\pi} (f_m(x))^2 dx = 0 \rightarrow \alpha_m = 0$ (3)

↳ (ex) sist. cu $\{\cos^{k_1}x, \cos^{k_2}x, \dots, \cos^{k_{m+1}}x\}$ este l. indep.

$k_1 < k_2 < \dots < k_{m+1}$

$\alpha_1 \cos^{k_1}x + \alpha_2 \cos^{k_2}x + \dots + \alpha_{m+1}\cos^{k_{m+1}}x = 0, \ \forall x$

Not. $\boxed{y = \cos x} \in [-1,1]$

$\Downarrow$

$\alpha_1 y^{k_1} + \alpha_2 y^{k_2} + \dots + \alpha_m y^{k_m} = 0, \ \forall y \in [-1,1]$

ecuație de grad k_m → trebuie să aibă o inf. de rădăcini

O ecuație de grad n are $(n+1)$ rădăcini. Cum arătăm că sunt distincte?

grad $f = n \rightarrow \{x_1, x_2, \dots, x_{n+1}\}$

$f(x) = (x - x_1)\, g(x)$

$f(x_2) = \underbrace{(x_2 - x_1)}_{\neq 0}\, g(x_2) \rightarrow x_2$ rad. $g(x_2)$

$\rightarrow x_3$ rad. $g(x_3)$

$\dots\dots\dots\dots$

$\rightarrow x_{n+1}$ rad. $g(x_{n+1})$

Prin inducție:

Grad 1 ... 2 rad

$ax + b = 0 \implies ax_1 + b = 0$

$ax_2 + b = 0$

$a(x_1 - x_2) = 0$

$\neq 0 \rightarrow a = 0$

$b = 0$

(ex) $V_{\mathbb{R}}$ $\{x, y, z\}$ l. i.

1) $\{x+y, y+z, z+x\}$ l. i.

2) $\{x-y, y-z, z-x\}$ l. d.

1) $\alpha_1(x+y) + \alpha_2(y+z) + \alpha_3(z+x) = 0$

$(\alpha_1 + \alpha_3)x + (\alpha_1 + \alpha_2)y + (\alpha_2 + \alpha_3)z = 0$ (4)

l. i.

$$\rightarrow \begin{cases} \alpha_1 + \alpha_3 = 0 \\ \alpha_1 + \alpha_2 = 0 \\ \alpha_2 + \alpha_3 = 0 \end{cases} \rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$$

$$2\alpha_1 + 2\alpha_2 + 2\alpha_3 = 0$$
$$\alpha_1 + \alpha_2 + \alpha_3 = 0$$

2) $(x-y)+(y-z) = x - z = -(z-x)$

$\rightarrow \underset{\alpha_1}{1}(x-y) + \underset{\alpha_2}{1}(y-z) + \underset{\alpha_3}{1}(z-x) = 0 \Rightarrow$ l. d.

(T) l. d. + sistem de generatori = BAZĂ

↳ $\mathbb{R}^3$: Baza canonică = $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$B = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\} \rightarrow$ bază pt $\mathbb{R}^3$

↳ $\mathbb{R}_2[X]$ B canonică $= \{1, X, X^2\}$

$B = \{-1+X, 2+2X, 3+X+2X^2\} \rightarrow$ bază pt $\mathbb{R}_2[X]$

(T) V_k sp. vectorial și are bază pe $\{v_1, v_2, \dots, v_n\}$, atunci

dim $V = n$

(T) $v \in V$

$v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$

$\Rightarrow \alpha_1, \alpha_2, \dots, \alpha_n$ se numesc coordonatele vectorului v în bază

exemplu $\mathbb{R}_2[X] \to \{-1+X, 1-X, 2+2X, 1+X^2\}$ $\begin{cases} s.g. \text{ pt. } \mathbb{R}_2[X] \\ l.d. \end{cases}$

$\mathbb{R}^3 \to \left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}1\\2\\1\end{pmatrix}\right\}$ $\begin{cases} l.i. \text{ în } \mathbb{R}^3 \\ \cancel{s.g.} \end{cases}$

(Ex) $\mathcal{B} = \left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}2\\-1\\2\end{pmatrix}, \begin{pmatrix}1\\2\\3\end{pmatrix}\right\}$ bază pt. $\mathbb{R}^3$ și det. coord. lui $x = \begin{pmatrix}6\\1\\9\end{pmatrix}$ în bază.

I. Arătăm liniara independență

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix} + \alpha_2\begin{pmatrix}2\\-1\\2\end{pmatrix} + \alpha_3\begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\Rightarrow \begin{cases} \alpha_1 + 2\alpha_2 + \alpha_3 = 0 \\ -\alpha_2 + 2\alpha_3 = 0 \Rightarrow \alpha_3 = \frac{1}{2}\alpha_2 \\ \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \end{cases}$$

Metoda lui Gauss: $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 2 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right) \xrightarrow{\alpha_3 - \alpha_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right) \uparrow$

$$\Rightarrow \left.\begin{cases} 2\alpha_3 = 0 \Rightarrow \alpha_3 = 0 \\ -\alpha_2 + 2\alpha_3 = 0 \Rightarrow \alpha_2 = 0 \\ \alpha_1 + 2\alpha_2 + \alpha_3 = 0 \Rightarrow \alpha_1 = 0 \end{cases}\right\} \Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$$

l. indep.

II. Sistem de generatori

$$\begin{pmatrix}a\\b\\c\end{pmatrix} = \alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix} + \alpha_2\begin{pmatrix}2\\-1\\2\end{pmatrix} + \alpha_3\begin{pmatrix}1\\2\\3\end{pmatrix}$$

$$\to \begin{cases} a = \alpha_1 + 2\alpha_2 + \alpha_3 \\ b = -\alpha_2 + 2\alpha_3 \\ c = \alpha_1 + 2\alpha_2 + \alpha_3 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & a \\ 0 & -1 & 2 & b \\ 1 & 2 & 3 & c \end{array}\right) \xrightarrow{L_3 - L_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & a \\ 0 & -1 & 2 & b \\ 0 & 0 & 2 & c-a \end{array}\right)$$

$\Rightarrow$ • $2\alpha_3 = c-a$

$$\boxed{\alpha_3 = \frac{c-a}{2}}$$

• $-\alpha_2 + 2\alpha_3 = b \Rightarrow \alpha_2 = -b + 2\alpha_3$

$$\boxed{\alpha_2 = -b + c - a}$$

• $\alpha_1 + 2\alpha_2 + \alpha_3 = a$

$$\Rightarrow \boxed{\alpha_1 = \frac{a-c}{2} + 2b - 2c + 2a + a}$$

III Determinarea coordonatelor

$$v = \begin{pmatrix} 6 \\ 1 \\ 9 \end{pmatrix} = ? \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + ? \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + ? \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

• $a = 6, b = 1, c = 9$

$\alpha_1 = \frac{1}{2}$; $\alpha_2 = 2$; $\alpha_3 = \frac{5}{2}$

subspații fundamentale

$A \in \mathcal{M}_{m \times n}(\mathbb{R})$

$A = (c_1 | c_2 | \dots | c_n)$, $c_i \in \mathbb{R}^m$

$\mathcal{C}(A) = \mathcal{I}(A) = \text{span}\{c_1, c_2, \dots, c_n\}$ → spațiul coloanelor

$\mathcal{N}(A) = \{X \in \mathbb{R}^n \mid AX = 0\}$

$\uparrow \mathbb{R}^n$

(Ex) $\mathcal{C}(A)$

$$A=\begin{pmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{pmatrix}$$

$$\mathcal{C}(A)=\left\{\begin{pmatrix}1\\2\\2\\-1\end{pmatrix},\begin{pmatrix}-3\\-6\\-6\\3\end{pmatrix},\begin{pmatrix}1\\9\\9\\-4\end{pmatrix},\begin{pmatrix}-2\\-1\\-1\\2\end{pmatrix},\begin{pmatrix}5\\8\\9\\-5\end{pmatrix},\begin{pmatrix}4\\2\\7\\-4\end{pmatrix}\right\} \rightarrow \text{vectori l. dep. (mâcar 2)}$$

$$\begin{pmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & 4 \end{pmatrix} \begin{matrix} \\ L_2-2L_1 \\ L_3-2L_1 \\ \longrightarrow \\ L_4+L_1 \end{matrix}$$

$$\rightarrow \begin{pmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 0 & 0 & 1 & 3 & -2 & -6 \\ 0 & 0 & 1 & 3 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ L_3-L_2 \\ \longrightarrow \\ \\ \end{matrix}$$

$$\rightarrow \begin{pmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 0 & 0 & 1 & 3 & -2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \mathcal{C}(A)=\text{span}\left\{\begin{pmatrix}1\\2\\2\\-1\end{pmatrix},\begin{pmatrix}1\\9\\9\\-4\end{pmatrix},\begin{pmatrix}5\\8\\9\\-5\end{pmatrix}\right\}$$

Subspații fundamentale

(Ex) $\mathcal{N}(B)=?$

$$B=\begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$$

$B \cdot X = 0$ (B → 3,5; X → 5,1)

$$\rightarrow \begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} -3x_1+6x_2-x_3+x_4-7x_5=0 \\ x_1-2x_2+2x_3+3x_4-x_5=0 \\ 2x_1-4x_2+5x_3+8x_4-4x_5=0 \end{cases}$$

$$\left(\begin{array}{ccccc|c} -3 & 6 & -1 & 1 & -7 & 0 \\ 1 & -2 & 2 & 3 & -1 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{array}\right) \begin{matrix} 3L_2+L_1 \\ \longrightarrow \\ 3L_3+2L_1 \end{matrix} \left(\begin{array}{ccccc|c} -3 & 6 & -1 & 1 & -7 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ 0 & 0 & 13 & 26 & -26 & 0 \end{array}\right) \begin{matrix} 5L_3-13L_2 \\ \longrightarrow \\ L_2/5 \end{matrix}$$

$$\longrightarrow \left(\begin{array}{ccccc|c} -3 & 6 & -1 & 1 & -7 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \longrightarrow \begin{cases} -3x_1+6x_2-x_3+x_4-7x_5=0 \\ x_3+2x_4-2x_5=0 \end{cases}$$

$$\Rightarrow \begin{matrix} x_1=2x_2+x_4-3x_5 \\ x_3=-2x_4+2x_5 \end{matrix}$$

$$\mathcal{N}(B)=\left\{ \begin{pmatrix} 2x_2+x_4-3x_5 \\ x_2 \\ -2x_4+2x_5 \\ x_4 \\ x_5 \end{pmatrix}, x_2,x_4,x_5 \in \mathbb{R} \right\}$$

$$= \left\{ x_2\begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_4\begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix} + x_5\begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix}, x_2,x_4,x_5 \in \mathbb{R} \right\}$$

$$\Rightarrow \text{Span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right\}$$ ↘ liniar indep. ↙ s. generatori → Bază pt. nucleu

(T) Dacă A este inversabilă $\Rightarrow \mathcal{N}(A)=\{0\}$

Sumă și intersecție de subspații

V; $U, W \subset V$

$$U+W=\{u+w \mid u \in U, w \in W\}$$

$V = U \oplus W \rightarrow \forall v \in V,\ \exists!\ u \in U$ și $w \in W$ a.î. $v = u + w$

(T) $\left.\begin{array}{l} U+W=V \\ U \cap W = \{0\} \end{array}\right| \Rightarrow V = U \oplus W$

$[\, v = U \oplus W \rightarrow \dim V = \dim U + \dim W$
$U \cap W = \{ x \mid x \in U$ și $x \in W \}\,]$

(T) Teorema Grassmann

$$\dim U + \dim W = \dim (U+W) + \dim (U \cap W)$$

(Ex) $\mathbb{R}^5$: $U_1 = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \middle| \ 2x_1 - x_2 - x_3 = 0,\ x_4 - 3x_5 = 0 \right\}$

$U_2 = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \middle| \ 2x_1 - x_2 + x_3 + 4x_4 + 4x_5 = 0 \right\}$

Găsiți suma și intersecția lor.

$$\begin{cases} 2x_1 - x_2 - x_3 = 0 \\ x_4 - 3x_5 = 0 \end{cases} \Rightarrow \begin{cases} x_2 = 2x_1 - x_3 \\ x_4 = 3x_5 \end{cases} \rightarrow U_1 = \left\{ \begin{pmatrix} x_1 \\ 2x_1 - x_3 \\ x_3 \\ 3x_5 \\ x_5 \end{pmatrix}, x_1, x_3, x_5 \in \mathbb{R} \right\}$$

$$= \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \\ 1 \end{pmatrix} \right\}$$

$$U_2 = \left\{ \begin{pmatrix} x_1 \\ 2x_1 + x_3 + 4x_4 + 4x_5 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \middle| \ x_1, x_3, x_4, x_5 \in \mathbb{R} \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

↳ $U_1+U_2 = \left\{ a\begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\-1\\1\\0\\0\end{pmatrix} + c\begin{pmatrix}0\\0\\0\\3\\1\end{pmatrix} + d\begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix} + e\begin{pmatrix}0\\1\\1\\0\\0\end{pmatrix} + f\begin{pmatrix}0\\4\\0\\0\\0\end{pmatrix} + g\begin{pmatrix}0\\4\\0\\0\\1\end{pmatrix} \right.$

$\left. a,b,c,d,e,f,g \in \mathbb{R} \right\}$

$= \underline{\text{span}}\left\{ \begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix}, \dots, \begin{pmatrix}0\\4\\0\\0\\1\end{pmatrix} \right\}$

Matriz: $\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & -1 & 0 & 2 & 1 & 4 & 4 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[3L_5-L_1]{L_2-2L_1}$

$\rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 4 & 4 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 3 \end{pmatrix} \xrightarrow{L_3+L_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 4 & 4 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 3 \end{pmatrix} \xrightarrow{L_3 \rightleftarrows L_4}$

$\rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 4 & 4 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & -1 & 3 \end{pmatrix}$ $\dim U_1+U_2 = \underline{\text{span}}\left\{ \begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix}, \begin{pmatrix}0\\-1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\0\\3\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\4\\0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\4\\0\\1\\0\end{pmatrix} \right\}$

↳ $U_2 \cap U_1 = ?$

$u \in U_1 \rightarrow u = a\begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\-1\\1\\0\\0\end{pmatrix} + c\begin{pmatrix}0\\0\\0\\3\\1\end{pmatrix}$

$u \in U_2 \rightarrow u = d\begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix} + e\begin{pmatrix}0\\1\\1\\0\\0\end{pmatrix} + f\begin{pmatrix}0\\4\\0\\1\\0\end{pmatrix} + g\begin{pmatrix}0\\4\\0\\0\\1\end{pmatrix}$ $\Big\} \rightarrow$

$$\rightarrow \begin{cases} a=d \\ 2a-b=2d+e+4f+4g \\ b=0 \\ 2c=f \\ c=g \end{cases} \rightarrow \begin{cases} a=d \\ b=e \\ e=-8g \rightarrow b=-8g \\ f=3c \\ c=g \end{cases} \rightarrow f=3g$$

$$u=d\begin{pmatrix}1\\2\\0\\0\end{pmatrix}+(-8g)\begin{pmatrix}0\\-1\\1\\0\end{pmatrix}+g\begin{pmatrix}0\\0\\0\\3\\1\end{pmatrix}=d\begin{pmatrix}1\\2\\0\\0\\0\end{pmatrix}+g\begin{pmatrix}0\\8\\-8\\3\\1\end{pmatrix}\rightarrow$$

$\rightarrow \dim U_1+\dim U_2=\dim(U_1+U_2)$

$+\dim(U_1\cap U_2)$

$3+4=5+2$ (adevărat)

Aplicații liniare

$T:U\to V$ s.n. **liniară dacă**:

1) $T(u+v)=T(u)+T(v)$
2) $T(\alpha u)=\alpha T(u)$

$\operatorname{Ker} T=\{u\in U \mid T(u)=0\}$ = nucleul

T- injectivă $\Leftrightarrow \operatorname{Ker} T=0$

T- surjectivă $\Leftrightarrow \operatorname{Im} T=V$

(T) Dacă d liniară este liniară și injectivă ($\dim\operatorname{Ker} T=0$)

$\{u_1,\dots,u_n\}$ l.i. din U

$\{T(u_1),T(u_2),\dots,T(u_n)\}$ l.i. din V

(T) $\dim \ker T + \dim \operatorname{Im} T = \dim$ (dimensiunea lui T)

↳ ! $A \in \mathcal{M}_{m\times n}(\mathbb{R})$, $T:\mathbb{R}^n \to \mathbb{R}^m$, $T(X) = A\cdot X$ $\Rightarrow$ $\ker(T) = \mathcal{N}(A)$, $\operatorname{Im}(T) = \mathcal{C}(A)$

↳ $T: U \to V$

$B_U = \{u_1, \dots, u_n\}$

$B_V = \{v_1, \dots, v_m\}$

$$\begin{cases} T(u_1) = a_{11}v_1 + a_{21}v_2 + \dots + a_{m1}x_m \\ T(u_2) = a_{12}v_1 + a_{22}v_2 + \dots + a_{m2}x_m \\ \vdots \\ T(u_n) = a_{1n}v_1 + a_{2n}v_2 + \dots + a_{nm}v_m \end{cases} \rightarrow A = (a_{ij})$$

matricea lui T relativă la bazele B_U și B_V

- $T: U \to V$ liniară

liniară ~ morfism

injectivă ~ monomorfism

surjectivă ~ epimorfism

$T: V \to V$ ~ endomorfism

bijectivă ~ izomorfism

(Ex) $T: \mathbb{R}_2[x] \to \mathbb{R}_2[x]$, $T(P) = 2P + (x+1)P'$

a) T. liniară

b) $\ker T = ?$, $\operatorname{Im} T = ?$

c) T inj.? T. surj.?

d) $\mathcal{M}(T, B_1, B_c)$

$B_1 = \{-1+x, 1+x, 1+x+x^2\}$

a) $T(P) = 2P + (X+1)P'$

① $T(P+Q) = 2(P+Q) + (X+1)(P+Q)' =$
$= 2P + 2Q + (X+1)P' + (X+1)Q' = T(P) + T(Q)$

② $T(\lambda P) = 2\lambda P + (X+1)(\lambda P)'$
$= \lambda [2P + (X+1)P']$

Cum ① și ② $\Rightarrow$ T - a. liniară

b) • $\operatorname{Ker} T = ?$

$P = aX^2 + bX + c$ / $T(P) = 0$ $\Rightarrow$ $2(aX^2+bX+c) + (X+1)(2aX+b) = 0$

$\rightarrow 4aX^2 + (3b+2a)X + (2c+b) = 0$

$\Rightarrow \begin{cases} a = 0; \\ b = 0; \\ c = 0. \end{cases}$ $\Rightarrow$ Singurul transf. T se află doar polinomul nul.

deci $\operatorname{Ker} T = 0$

• $\operatorname{Im} T = ?$

$\operatorname{Im}(T) = \{T(P) \mid P \in \mathbb{R}_2[X]\}$

$= \{4aX^2 + (3b+2a)X + 2c + b \mid a_i, b_i, c \in \mathbb{R}\}$

$= \{a(4X^2+2X) + b(3X+1) + 2c \cdot 1 \mid a, b, c \in \mathbb{R}\}$

$= \operatorname{span}\{4X^2+2X,\ 3X+1,\ 1\} \Rightarrow$ deci $\dim \operatorname{Im} T = 3$

c) deci $\operatorname{Ker} T = 0 \rightarrow T$: injectivă.
deci $\operatorname{Im} T = 3$ (deci. domeniului) $\rightarrow T$: surjectivă

Dacă deci $\operatorname{Ker} T =$ deci $\operatorname{Im} T \Rightarrow$ coincid.

d) $T(-1+X) = -2 + 2X + (X+1)\cdot 1 = 1 \cdot X^0 + 3 \cdot X^1 + 0 \cdot X^2$

$T(1+X) = 2 + 2X + 1 \cdot (X+1) = 3 + 3X = 3 \cdot X^0 + 3 \cdot X^1 + 0 \cdot X^3$

$M = \begin{pmatrix} -1 & 3 & 3 \\ 3 & 3 & 5 \\ 0 & 0 & 4 \end{pmatrix}$ matricea de trecere dintr-o bază oarecare în $\mathcal{B}_1$.

Spații vectoriale cu produs scalar.

$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ s.n. produs scalar dacă

1) $\langle u, u \rangle \geq 0 \quad (=0 \Leftrightarrow u = 0)$
2) $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$
3) $\langle u, \alpha v \rangle = \alpha \langle u, v \rangle$
4) $\langle u, v \rangle = \langle v, u \rangle$

Exemplu $\mathbb{R}^n \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \; y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$\langle x, y \rangle = x^T y = x_1 y_1 + \dots + x_n y_n$

Exemplu $C([a,b])$ (sau $L^2(a,b)$)

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x) \, dx$$

Norma: $\|u\| = \sqrt{\langle u, u \rangle}$

$\|\lambda u\| = |\lambda| \cdot \|u\|$

$\|u+v\| \leq \|u\| + \|v\|$ (inegalitatea Δ)

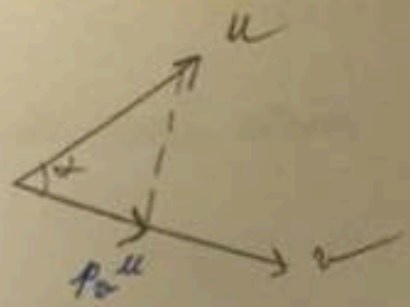
$\mathbb{R}^n \quad \|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

Inegalitatea Cauchy-Schwarz

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$

$$\left|\underbrace{\frac{\langle u, v\rangle}{\|u\| \cdot \|v\|}}_{\cos\alpha}\right| \leq 1.$$

$$\Rightarrow \cos\alpha = \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|}$$



$$pr_v u = \|pr_v u\| \cdot versor$$

$$pr_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} \cdot v$$

Două vectori formează $\left(\alpha = \frac{\pi}{2}\right) \rightarrow u \perp v \Rightarrow \langle u, v\rangle = 0$

Complement ortogonal: $U^\perp = \{v \in V \mid \langle v, u\rangle = 0, \forall u \in U\}$

(Ex) Găsiți compl. ortogonal

$$U = span\left\{\begin{pmatrix}1\\1\\3\\0\end{pmatrix}, \begin{pmatrix}2\\-2\\0\\5\end{pmatrix}\right\} \subset \mathbb{R}^4$$

Fie $\begin{pmatrix}a\\b\\c\\d\end{pmatrix} \in \mathbb{R}^4$ a.î. $\begin{pmatrix}a\\b\\c\\d\end{pmatrix} \perp \begin{pmatrix}1\\1\\3\\0\end{pmatrix} \rightarrow a+b+3c=0 \Rightarrow a=-b-3c$

a.î. $\begin{pmatrix}a\\b\\c\\d\end{pmatrix} \perp \begin{pmatrix}2\\-2\\0\\5\end{pmatrix} \rightarrow 2a-2b+5d=0$

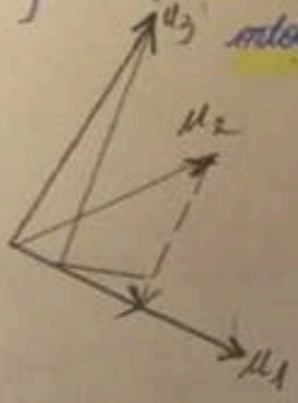
$$\rightarrow d = \frac{4b+6c}{5}$$

$$U^\perp = \left\{\begin{pmatrix}-b-3c\\b\\c\\\frac{4}{5}b+\frac{6}{5}c\end{pmatrix}\right\} = span\left\{\begin{pmatrix}-1\\1\\0\\\frac{4}{5}\end{pmatrix}, \begin{pmatrix}-3\\0\\1\\\frac{6}{5}\end{pmatrix}\right\}$$

Procedeul de ortonormalizare Gram Schmidt

$\{u_1, u_2, \dots, u_n\} \to \{v_1, v_2, \dots, v_n\} \to \{w_1, w_2, \dots, w_n\}$

oarecare — ortonormală



- $v_1 = u_1$

$v_2 = u_2 - pr_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} \cdot v_1$

$v_3 = u_3 - pr_{v_2} u_3 = u_3 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} \cdot v_2$

(Ex) Fol. proc. de ortonorm. G. Schm., ortog. urm. bază a lui $\mathbb{R}^3$

$B = \left\{ \underbrace{\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{u_2}, \underbrace{\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}}_{u_3} \right\}$

$v_1 = u_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$v_2 = u_2 - proj_{v_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} \cdot v_1$

$= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 2/3 \\ 1/3 \end{pmatrix}$

$v_3 = u_3 - proj_{v_2} u_3 = u_3 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} \cdot v_2$

$= \begin{pmatrix} -1/2 \\ 0 \\ 1/2 \end{pmatrix}$

$$w_1 = \frac{1}{\|v_1\|} \cdot v_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$w_2 = \frac{1}{\|v_2\|} \cdot v_2 = \frac{\sqrt{3}}{\sqrt{2}} \begin{pmatrix} 1/3 \\ 2/3 \\ 1/3 \end{pmatrix}$$

$$w_3 = \frac{1}{\|v_3\|} \cdot v_3 = v_2 \begin{pmatrix} -1/2 \\ 0 \\ 1/2 \end{pmatrix}$$

(ex) Fie. $A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \\ \smile & \smile & \smile \\ u_1 & u_2 & u_3 \end{pmatrix}$ Găsiți descompunerea QR.

u_1, u_2, u_3 – l. indep. $\xrightarrow{G\text{-}S} \{w_1, w_2, w_3\}$ ortonormată.

$$A = \underbrace{(w_1 | w_2 | \cdots | w_n)}_{Q}$$

(ortogonală, $Q^T = Q^{-1}$)

$$\underbrace{\begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_2, u_2 \rangle & \cdots & \langle w_1, u_n \rangle \\ 0 & \langle w_2, u_2 \rangle & \cdots & \langle w_2, u_n \rangle \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \langle w_n, u_n \rangle \end{pmatrix}}_{R}$$

$$Q = \begin{pmatrix} 1/\sqrt{3} & \sqrt{6}/6 & -\sqrt{2}/2 \\ -1/\sqrt{3} & \sqrt{6}/3 & 0 \\ 1/\sqrt{3} & \sqrt{6}/6 & \sqrt{2}/2 \end{pmatrix}$$

w_1 w_2 w_3

$$R = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{\sqrt{6}}{3} & \frac{2}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & 5\sqrt{6}/6 \\ 0 & 0 & \sqrt{2}/2 \end{pmatrix}$$

$Q \cdot R = A$

Valori și vectori proprii

$A \in \mathcal{M}_n(\mathbb{R})$

$A \cdot v = \lambda \cdot v$ unde $v \in \mathbb{R}^4$, $\lambda \in \mathbb{R}$

(T) Dacă λ-v. proprie $\Leftrightarrow$ $\det(A - \lambda I_n) = 0$ → matrice neinversabilă

(T) $\lambda_1 + \lambda_2 + \dots + \lambda_n = tr(A)$

O matrice este SIMETRICĂ dacă $A^T = A$

→ 1) $\forall$ v p $\in \mathbb{R}$

→ 2) vectorii p. ai op. val. pr. ≠ sunt ortogonali

(Ex) $A = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}$, $\det(A - \lambda I_3) = \det \begin{pmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{pmatrix} = 0 \rightarrow$

$\rightarrow (3-\lambda)(-\lambda)(3-\lambda) + 16 + 16\lambda - 4(3-\lambda) - 4(3-\lambda) = 0 \rightarrow$

$\rightarrow -9\lambda + 3\lambda^2 + 3\lambda^2 - \lambda^3 + 32 + 16\lambda - 12 + 4\lambda - 12 + 4\lambda = 0$

$\rightarrow -\lambda^3 + 6\lambda^2 + 15\lambda + 8 = 0 \rightarrow \lambda^3 - 6\lambda^2 - 15\lambda - 8 = 0 \rightarrow$ ↓div. lui 8

$\rightarrow \lambda^2(\lambda+1) - 7\lambda(\lambda+1) - 8(\lambda+1) = 0 \rightarrow$

$\rightarrow (\lambda+1)(\lambda^2 - 7\lambda - 8) = 0 \rightarrow (\lambda+1)^2(\lambda - 8) = 0$

$\Rightarrow \lambda_{1,2} = -1$

$\lambda_3 = 8$

$\lambda_{-1} \rightarrow AV = \lambda V \rightarrow AV = -V \Rightarrow \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = -\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \rightarrow$

$$\begin{cases} 4v_1 + 2v_2 + 4v_3 = 0 \\ 2v_1 + v_2 + 2v_3 = \\ 4v_1 + 2v_2 + 4v_3 = 0 \end{cases} \rightarrow v_2 = -2v_1 - 2v_3 \Rightarrow V_{-1} = \left\{ \begin{pmatrix} v_1 \\ -2v_1 - 2v_3 \\ v_3 \end{pmatrix} \right\}$$

$$= Sp\left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \right\}$$

$\lambda_3 \rightarrow A \cdot V = \lambda V \rightarrow A \cdot V = 8V \Rightarrow \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 8 \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

$$\rightarrow \begin{cases} -5v_1 + 2v_2 + 4v_3 = 0 \\ 2v_1 - 8v_2 + 2v_3 = 0 \\ 4v_1 + 2v_2 - 5v_3 = 0 \end{cases} \rightarrow \begin{cases} -5v_1 + 2v_2 = -4v_3 \\ 4v_1 + 2v_2 = +5v_3 \end{cases}$$

$\Rightarrow \boxed{v_1 = v_3}$

$\boxed{v_3 = 2v_2}$

$$\Rightarrow \left\{ \begin{pmatrix} 2v_2 \\ v_2 \\ 2v_2 \end{pmatrix} \right\} = Sp\left\{ \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \right\}$$

$\begin{cases} \dim V_{-1} = 2 \; ; \; m_{(-1)} = 2 \\ \dim V_3 = 1 \; ; \; m_{(8)} = 1 \end{cases}$ Dacă m (mult. alg.) coincide cu dim (mult. geometrică) $\Rightarrow$ matrica este diagonalizabilă.

a.î. $\boxed{A = S D \cdot S^{-1}}$

D - matricea formată din val. prop. pe diagonală

$$D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 8 \end{pmatrix} \quad \text{și} \quad S = \begin{pmatrix} 1 & 0 & 2 \\ -2 & -2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$A = S \cdot D \cdot S^{-1} \rightarrow A^2 = A \cdot A = SD \cdot S^{-1} \cdot S \cdot D \cdot S^{-1}$

Prin inducție se ajunge la → $\boxed{A^n = S \cdot D^n \cdot S^{-1}}$

(Ex) $(x_n)_{n \geq 1}$ $x_{n+2} = 6x_{n+1} - 5x_n$, $x_1 = 1, x_2 = 2$

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \begin{pmatrix} 6x_{n+1} - 5x_n \\ x_n \end{pmatrix} = \underbrace{\begin{pmatrix} 6 & -5 \\ 1 & 0 \end{pmatrix}}_{A} \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix}$$

prin recurență
$$= A^{n-1} \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}.$$

$$A = SDS^{-1} \Rightarrow A^{n-1} = SD^{n-1}S^{-1}$$

$$\| $$

$$\begin{pmatrix} \lambda_1^{n-1} & 0 & \cdots \\ 0 & \lambda_2^{n-1} & 0 \\ \vdots & 0 & \ddots \\ 0 & 0 & \lambda_m^{n-1} \end{pmatrix}$$

Ridicarea la putere specificată a elementelor de pe diagonala principală